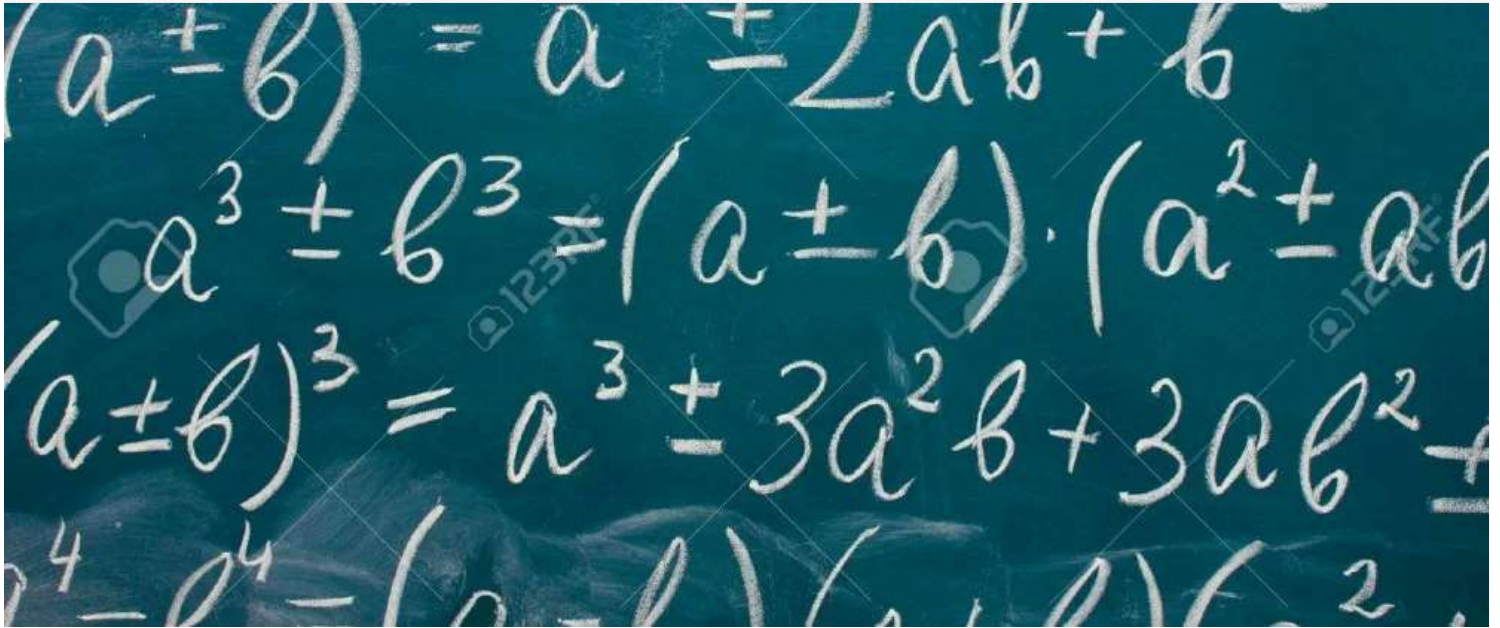


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Back of the Envelope Calculation

#interview #systemdesign

Intro

I found this topic rather misleading and always overcomplicated. Though I cannot disagree the version below is a lot more simplified than real life calculations, it's still covers 99% of the things you can encounter in your interview process.

What to estimate?

QPS - queries per second

RPS - reads per second

WPS - writes per second

Peak QPS = QPS * 2 (usually)

RW - read write ratio

Message size - size of the message in bytes if not given

Read Throughput - $\text{RPS} * \text{message size} = \text{N bytes per second}$

Write Throughput - $\text{WPS} * \text{message size} = \text{N bytes per second}$

💡 **Throughput** is how much data actually passed through and **bandwidth** is how much data CAN be passed through (network configuration)

Ex: 1gbps network **bandwidth** can pass 125mb/s

Storage - usually storage for N years

Replica storage - storage * 2-3 times

Cache storage - usually 20% of storage or so

Cache replica storage - cache storage * 2-3 times

Basic Numbers

seconds in a day - $24 * 60 * 60 = 86400$, roughly 10^5

1 ASCII letter - 1 char

timestamp - 8 bytes (2^{64})

10^3 - 1kb

10^6 - 1mb

10^9 - 1gb

10^{12} - 1tb

10^{15} - 1pb

10^{18} - 1eb

Powers of two

Power	Exact Value	Approx Value	Bytes
7	128		
8	256		
10	1024	1 thousand	1 KB
16	65,536		64 KB
20	1,048,576	1 million	1 MB
30	1,073,741,824	1 billion	1 GB
32	4,294,967,296		4 GB
40	1,099,511,627,776	1 trillion	1 TB

Latency numbers every programmer should know

Latency Comparison Numbers					

L1 cache reference	0.5	ns			
Branch mispredict	5	ns			
L2 cache reference	7	ns			14x L1 cache
Mutex lock/unlock	25	ns			
Main memory reference	100	ns			20x L2 cache,
Compress 1K bytes with Zippy	10,000	ns	10	us	
Send 1 KB bytes over 1 Gbps network	10,000	ns	10	us	
Read 4 KB randomly from SSD*	150,000	ns	150	us	~1GB/sec SSD
Read 1 MB sequentially from memory	250,000	ns	250	us	
Round trip within same datacenter	500,000	ns	500	us	
Read 1 MB sequentially from SSD*	1,000,000	ns	1,000	us	1 ms ~1GB/sec SSD,
HDD seek	10,000,000	ns	10,000	us	10 ms 20x datacenter
Read 1 MB sequentially from 1 Gbps	10,000,000	ns	10,000	us	10 ms 40x memory, 16
Read 1 MB sequentially from HDD	30,000,000	ns	30,000	us	30 ms 120x memory, 36
Send packet CA->Netherlands->CA	150,000,000	ns	150,000	us	150 ms
Notes					

1 ns = 10 ⁻⁹ seconds					
1 us = 10 ⁻⁶ seconds = 1,000 ns					
1 ms = 10 ⁻³ seconds = 1,000 us = 1,000,000 ns					

Handy metrics based on latency numbers

Read sequentially from HDD at 30 MB/s
Read sequentially from 1 Gbps Ethernet at 100 MB/s
Read sequentially from SSD at 1 GB/s
Read sequentially from main memory at 4 GB/s

6-7 world-wide round trips per second
2,000 round trips per second within a data center

How to estimate?

1. Clarify number of daily users and number of total users.
2. Ask about number of request from user on average. From here you can get QPS.
3. Think about peak QPS, Reads and Writes.
4. Assume (clarify) message size.
5. Calculate throughput.
6. If it's possible think about average data size. And calculate storage and cache here.

Estimation example

You have 10M daily active users and each of them makes 100 read requests per day on average and new data is created 5 times per day.

$$\text{RPS} = 10\text{M} * 100 / 86400 = 12000 \text{ r/s}$$

$$\text{WPS} = 10\text{M} * 5 / 86400 = 580 \text{ w/s}$$

$$\text{Peak QPS} = 24000 \text{ r/s}$$

Let's assume (clarify with the interviewer) that the average read message size is 50 bytes and the written message is 1kb.

$$\text{Avg Read throughput} = 50 * 12 * 10^3 = 60\text{kb/s}$$

$$\text{Avg Write throughput} = 1\text{kb} * 580 = 580\text{kb/s}$$

Here we can think about the type of data/metadata etc. Let's assume that you have clarified with your interviewer and the size of the new data is 1kb.

$$\text{5 years storage} = 10\text{M} * 1\text{kb} * 5 \text{ time per day} * 365 \text{ days per year} * 5 \text{ years} = 91\text{tb} * 3 = 300\text{tb with replicas.}$$

Lets assume that you have only 10% of hot data and you agreed to use 20% as cache.

$$\text{Cache storage} = 10\% * 90\text{tb} * 20\% * 3 \text{ replicas} = 5.5 \text{ tb}$$